



Some thesis title

by

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Someone.

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* Supervisor X.

* Funding Y.

“ *A very long quote*
from someone very important. ”

– Mr M

Thank you to Leanne for helping to prepare this template.

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ABSTRACT

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NOMENCLATURE

ACRONYMS AND ABBREVIATIONS

MFT My Funky Thesis

DEFINITIONS

S

STUDENT is an entity needing a thesis to transcend the state of being a student.

INTRODUCTION

1.1 BACKGROUND

Since the late 1990s there have been indications that whales have been struck by shipping vessels.[1] Today this has become a larger concern, as vessel speeds and traffic has substantially increased. In False Bay, South Africa, a colony of Bryde's whales have been detected. Bryde's whales are found to migrate in smaller regions instead of the usual large-scale migrations of other whale species. Due to the increase in shipping traffic as well as the fishing activities in the area, the Bryde's whales in this area are also in danger.[2] Detection and classification systems for whale calls can be used to monitor the migration patterns of the specific whale colony throughout the year. The information gained from such analysis can help determine when vessel traffic should be diverted for the safety of the whale population. These types of systems have been implemented on different whale colonies, but the migration patterns of the False Bay colony has not yet been thoroughly documented.[3] Due to the variations in call types between different colonies of some whale species (killer whales[4], blue whales[5]), models trained for other Bryde's whale colonies would not be completely trustworthy for the detection of calls within the False Bay Bryde's Whale colony as universality of calls cannot be assumed. Therefore, a model trained specifically for the False Bay colony would be beneficial. Dynamic Time Warping (DTW) is a lesser studied method for whale classification, but has been shown to be effective on different whale colonies.[6]

1.2 PROBLEM STATEMENT

This project will determine whether using Dynamic Time Warping (DTW) in the frequency domain, thus using vector features, is able to effectively detect Bryde's Whale calls within the False Bay region. Three feature extraction methods will be compared. These will be

Short Time Fourier Transform (STFT), Mel-frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC).

1.3 OBJECTIVES

1. Implement a DTW detection system using three different feature extraction methods(STFT, MFCC and LPC). This shows whether the effectiveness of the DTW system is independent from the feature extraction method.
2. Calculate the precision, recall and F1-score for each model.
3. Compare results and determine whether DTW is overall a good detection method and which feature extraction method is best suited.

1.4 SCOPE

What you didn't consider.

1.5 LAYOUT OF REPORT

- * Chapter 2 provides any background information needed to understand how to solve the given problem as well as any helpful literature of previous solutions to the problem.
- * Chapter 3 provides an explanation of the overall setup and method used to solve the problem including what metrics were used to meet the stated objectives.
- * Chapter 4 gives a detailed design explanation for all the functional blocks defined in the previous chapter.
- * Chapter 5 provides evidence of results obtained after the system has been designed as well as descriptions of what the results show.
- * Chapter 6 contains all concluding thoughts on the results and what that means for the solutions as well as improvements and recommendations for the project.

My funky thesis (MFT) is as good as MFT can be, as Table 1.1 and Figure 1.1 shows. Have a look at my explanation in Section 2.1 on page 4, which is in Chapter 2.

Table 1.1: Your table caption

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Description	100	140	24601

WELCOME TO
ENGINEERING

Figure 1.1: Your figure caption.

LITERATURE

2.1 RELATED WORK

SYSTEM DESIGN

DETAILED DESIGN

RESULTS

CONCLUSION

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APPENDIX A

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